

```
[nivass@localhost ~]$ ls
a b c Desktop Documents Downloads Music Pictures Public rec Templates Videos
[nivass@localhost ~]$ cd a
[nivass@localhost a]$ ls
'\ a b c d e f os
[nivass@localhost a]$ mkdir g h i
[nivass@localhost a]$ ls
'\ a b c d e f g h i os
[nivass@localhost a]$ mkdir -p x/y/z
[nivass@localhost a]$ cd x
[nivass@localhost x]$ cd y
[nivass@localhost y]$ cd z
[nivass@localhost z]$ cd ..
[nivass@localhost y]$ cd ..
[nivass@localhost x]$ cd ..
[nivass@localhost a]$ cd x
[nivass@localhost x]$ cd y
[nivass@localhost y]$ rmdir z
[nivass@localhost y]$ ls
[nivass@localhost y]$ cd ..
[nivass@localhost x]$ rmdir y
[nivass@localhost x]$ cd ..
[nivass@localhost a]$ rmdir x
[nivass@localhost a]$ ls
'\ a b c d e f g h i os
[nivass@localhost a]$ mkdir -p x/y/z
[nivass@localhost a]$ rmdir -rf x
rmdir: invalid option -- 'r'
Try 'rmdir --help' for more information.
[nivass@localhost a]$ rm -rf x
[nivass@localhost a]$ ls
'\ a b c d e f g h i os
[nivass@localhost a]$ cat > os
^Z
[1]+  Stopped                  cat > os
"Hi Hello . This is OS."
quit
^C
```



```
[nivass@localhost ~]$ cd rec
[nivass@localhost rec]$ vi rec1
[nivass@localhost rec]$ ls
rec1  recnotes
[nivass@localhost rec]$ cat recnotes
np cmd is used to copy the files in one folder to the another folder
mv cmd is used to move the file from one folder to the another folder
hi world!!!
[nivass@localhost rec]$ touch rec1
[nivass@localhost rec]$ ls
rec1  recnotes
[nivass@localhost rec]$ cp rec1 recnotes
[nivass@localhost rec]$ cat recnotes
np cmd is used to copy the files in one folder to the another folder
mv cmd is used to move the file from one folder to the another folder
hi world!!!
[nivass@localhost rec]$ mv recnotes rec1
[nivass@localhost rec]$ ls
rec1
[nivass@localhost rec]$ cat rec1
np cmd is used to copy the files in one folder to the another folder
mv cmd is used to move the file from one folder to the another folder
hi world!!!
[nivass@localhost rec]$ date
Wednesday 21 January 2026 09:49:58 PM IST
[nivass@localhost rec]$ date +%d
21
[nivass@localhost rec]$ date +%m
01
[nivass@localhost rec]$ date +%h
Jan
[nivass@localhost rec]$ date +%y
26
[nivass@localhost rec]$ date +%H
21
[nivass@localhost rec]$ date +%M
51
[nivass@localhost rec]$ date +%S
16
[nivass@localhost rec]$
```


16

[nivass@localhost rec]\$ cal 2026

2026

January							February							March						
Su	Mo	Tu	We	Th	Fr	Sa	Su	Mo	Tu	We	Th	Fr	Sa	Su	Mo	Tu	We	Th	Fr	Sa
				1	2	3	1	2	3	4	5	6	7	1	2	3	4	5	6	7
4	5	6	7	8	9	10	8	9	10	11	12	13	14	8	9	10	11	12	13	14
11	12	13	14	15	16	17	15	16	17	18	19	20	21	15	16	17	18	19	20	21
18	19	20	21	22	23	24	22	23	24	25	26	27	28	22	23	24	25	26	27	28
25	26	27	28	29	30	31								29	30	31				

April							May							June							
Su	Mo	Tu	We	Th	Fr	Sa	Su	Mo	Tu	We	Th	Fr	Sa	Su	Mo	Tu	We	Th	Fr	Sa	
			1	2	3	4						1	2			1	2	3	4	5	6
5	6	7	8	9	10	11	3	4	5	6	7	8	9	7	8	9	10	11	12	13	
12	13	14	15	16	17	18	10	11	12	13	14	15	16	14	15	16	17	18	19	20	
19	20	21	22	23	24	25	17	18	19	20	21	22	23	21	22	23	24	25	26	27	
26	27	28	29	30			24	25	26	27	28	29	30	28	29	30					
							31														

July							August							September						
Su	Mo	Tu	We	Th	Fr	Sa	Su	Mo	Tu	We	Th	Fr	Sa	Su	Mo	Tu	We	Th	Fr	Sa
			1	2	3	4							1			1	2	3	4	5
5	6	7	8	9	10	11	2	3	4	5	6	7	8	6	7	8	9	10	11	12
12	13	14	15	16	17	18	9	10	11	12	13	14	15	13	14	15	16	17	18	19
19	20	21	22	23	24	25	16	17	18	19	20	21	22	20	21	22	23	24	25	26
26	27	28	29	30	31		23	24	25	26	27	28	29	27	28	29	30			
							30	31												

October							November							December						
Su	Mo	Tu	We	Th	Fr	Sa	Su	Mo	Tu	We	Th	Fr	Sa	Su	Mo	Tu	We	Th	Fr	Sa
				1	2	3	1	2	3	4	5	6	7			1	2	3	4	5
4	5	6	7	8	9	10	8	9	10	11	12	13	14	6	7	8	9	10	11	12
11	12	13	14	15	16	17	15	16	17	18	19	20	21	13	14	15	16	17	18	19
18	19	20	21	22	23	24	22	23	24	25	26	27	28	20	21	22	23	24	25	26
25	26	27	28	29	30	31	29	30						27	28	29	30	31		

[nivass@localhost rec]\$ cal 01 2026

January 2026

Su	Mo	Tu	We	Th	Fr	Sa
			1	2	3	
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

[nivass@localhost rec]\$ █

18	19	20	21	22	23	24	22	23	24	25	26	27	28	20	21	22	23	24	25	26
25	26	27	28	29	30	31	29	30	27	28	29	30	31							

```
[nivass@localhost rec]$ cal 01 2026
```

```
January 2026
Su Mo Tu We Th Fr Sa
                1  2  3
 4  5  6  7  8  9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30 31
```

```
[nivass@localhost rec]$ echo os
```

```
os
```

```
[nivass@localhost rec]$ echo ""
```

```
[nivass@localhost rec]$ echo "hi hello"
```

```
hi hello
```

```
[nivass@localhost rec]$ bc -l
```

```
bc 1.07.1
```

```
Copyright 1991-1994, 1997, 1998, 2000, 2004, 2006, 2008, 2012-2017 Free Software Foundation, Inc.
```

```
This is free software with ABSOLUTELY NO WARRANTY.
```

```
For details type 'warranty'.
```

```
100-2
```

```
98
```

```
200/2
```

```
100.000000000000000000000000
```

```
quit
```

```
[nivass@localhost rec]$ who
```

```
nivass  tty2          2026-01-21 21:45 (local)
```

```
[nivass@localhost rec]$ whoami
```

```
nivass
```

```
[nivass@localhost rec]$ id
```

```
uid=1000(nivass) gid=1000(nivass) groups=1000(nivass) context=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
```

```
[nivass@localhost rec]$ tty
```

```
/dev/pts/0
```

```
[nivass@localhost rec]$ man ls
```

```
[nivass@localhost rec]$ man cd
```

```
[nivass@localhost rec]$ ps
```

```
PID TTY          TIME CMD
```

```
2997 pts/0      00:00:00 bash
```

```
3283 pts/0      00:00:00 ps
```

```
[nivass@localhost rec]$ uname -a
```

```
Linux localhost.localdomain 5.14.0-658.el9.x86_64 #1 SMP PREEMPT_DYNAMIC Fri Jan 9 20:29:02 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux
```

```
[nivass@localhost rec]$ █
```


NAME

ls - list directory contents

SYNOPSIS

ls [OPTION]... [FILE]...

DESCRIPTION

List information about the FILES (the current directory by default). Sort entries alphabetically if none of **-cftuvSUX** nor **--sort** is specified. Mandatory arguments to long options are mandatory for short options too.

-a, --all

do not ignore entries starting with .

-A, --almost-all

do not list implied . and ..

--author

with **-l**, print the author of each file

-b, --escape

print C-style escapes for nongraphic characters

--block-size=SIZE

with **-l**, scale sizes by SIZE when printing them; e.g., **'--block-size=M'**; see SIZE format below

-B, --ignore-backups

do not list implied entries ending with ~

-c with **-lt**: sort by, and show, ctime (time of last modification of file status information); with **-l**: show ctime and sort by name; otherwise: sort by ctime, newest first

-C list entries by columns

--color[=WHEN]

colorize the output; WHEN can be 'always' (default if omitted), 'auto', or 'never'; more info below

-d, --directory

list directories themselves, not their contents

-D, --dired

generate output designed for Emacs' dired mode

NAME

bash, :, ., [, alias, bg, bind, break, builtin, caller, cd, command, compgen, complete, compopt, continue, declare, dirs, disown, echo, enable, eval, exec, exit, export, false, fc, fg, getopts, hash, help, history, jobs, kill, let, local, logout, mapfile, popd, printf, pushd, pwd, read, readonly, return, set, shift, shopt, source, suspend, test, times, trap, true, type, typeset, ulimit, umask, unalias, unset, wait - bash built-in commands, see **bash(1)**

BASH BUILTIN COMMANDS

Unless otherwise noted, each builtin command documented in this section as accepting options preceded by - accepts -- to signify the end of the options. The :, true, false, and test/[builtins do not accept options and do not treat -- specially. The exit, logout, return, break, continue, let, and shift builtins accept and process arguments beginning with - without requiring --. Other builtins that accept arguments but are not specified as accepting options interpret arguments beginning with - as invalid options and require -- to prevent this interpretation.

: [arguments]

No effect; the command does nothing beyond expanding arguments and performing any specified redirections. The return status is zero.

. filename [arguments]

source filename [arguments]

Read and execute commands from filename in the current shell environment and return the exit status of the last command executed from filename. If filename does not contain a slash, filenames in **PATH** are used to find the directory containing filename. The file searched for in **PATH** need not be executable. When **bash** is not in posix mode, the current directory is searched if no file is found in **PATH**. If the **sourcepath** option to the **shopt** builtin command is turned off, the **PATH** is not searched. If any arguments are supplied, they become the positional parameters when filename is executed. Otherwise the positional parameters are unchanged. If the -T option is enabled, **source** inherits any trap on **DEBUG**; if it is not, any **DEBUG** trap string is saved and restored around the call to **source**, and **source** unsets the **DEBUG** trap while it executes. If -T is not set, and the sourced file changes the **DEBUG** trap, the new value is retained when **source** completes. The return status is the status of the last command exited within the script (0 if no commands are executed), and false if filename is not found or cannot be read.

alias [-p] [name=value] ...]

Alias with no arguments or with the -p option prints the list of aliases in the form **alias name=value** on standard output. When arguments are supplied, an alias is defined for each name whose value is given. A trailing space in value causes the next word to be checked for alias substitution when the alias is expanded. For each name in the argument list for which no value is supplied, the name and value of the alias is printed. **Alias** returns true unless a name is given for which no alias has been defined.

bg [jobspec ...]

Resume each suspended job jobspec in the background, as if it had been started with &. If jobspec is not present, the shell's notion of the current job is used. **bg jobspec** returns 0 unless run when job control is disabled or, when run with job control enabled, any specified job-spec was not found or was started without job control.

bind [-m keymap] [-lpsvPSVX]

bind [-m keymap] [-q function] [-u function] [-r keyseq]

bind [-m keymap] -f filename

bind [-m keymap] -x keyseq:shell-command

bind [-m keymap] keyseq:function-name

bind [-m keymap] keyseq:readline-command


```
[nivass@localhost ~]$ ls
a b c Desktop Documents Downloads Music Pictures Public rec Templates Videos
[nivass@localhost ~]$ rec
bash: rec: command not found...
[nivass@localhost ~]$ cd rec
[nivass@localhost rec]$ ls
os rec1 rec2 renotes
[nivass@localhost rec]$ wc os
1 5 47 os
[nivass@localhost rec]$ who | tee os | wc -l
1
[nivass@localhost rec]$ cat os
nivass    tty2          2026-01-23 08:27 (local)
[nivass@localhost rec]$ cp renotes os
[nivass@localhost rec]$ cat os
[nivass@localhost rec]$ vi os
[nivass@localhost rec]$ cat os
"Hi hello OS."
ok it's working.
[nivass@localhost rec]$ sort os
"Hi hello OS."
ok it's working.
[nivass@localhost rec]$ sort -r os
ok it's working.
"Hi hello OS."
[nivass@localhost rec]$ nl os
1 "Hi hello OS."
2 ok it's working.
[nivass@localhost rec]$ grep "Os" os
[nivass@localhost rec]$ grep "o" od
grep: od: No such file or directory
[nivass@localhost rec]$ grep "o" os
```



```
[nivass@localhost rec]$ grep "o" os
"Hi hello o OS."
ok it's working.
[nivass@localhost rec]$ ls | more
os
rec1
rec2
recnotes
[nivass@localhost rec]$ file os
os: ASCII text
[nivass@localhost rec]$ head os
"Hi hello OS."
ok it's working.
[nivass@localhost rec]$ tail os
"Hi hello OS."
ok it's working.
[nivass@localhost rec]$ head -1 os
"Hi hello OS."
[nivass@localhost rec]$ tail -1 os
ok it's working.
[nivass@localhost rec]$ sort -c os
[nivass@localhost rec]$ sort -n os
"Hi hello OS."
ok it's working.
[nivass@localhost rec]$ sort -m os
"Hi hello OS."
ok it's working.
[nivass@localhost rec]$ cut -c 1-5 os
"Hi h
ok i
[nivass@localhost rec]$ free
```

	total	used	free	shared	buff/cache	available
--	-------	------	------	--------	------------	-----------

"Hi h

ok i

[nivass@localhost rec]\$ free

	total	used	free	shared	buff/cache	available
Mem:	1746764	940540	337480	20224	647648	806224
Swap:	2097148	0	2097148			

[nivass@localhost rec]\$ free -h

	total	used	free	shared	buff/cache	available
Mem:	1.7Gi	918Mi	329Mi	19Mi	632Mi	787Mi
Swap:	2.0Gi	0B	2.0Gi			

[nivass@localhost rec]\$ top

top - 08:52:42 up 25 min, 1 user, load average: 0.00, 0.04, 0.06

Tasks: **273** total, **1** running, **272** sleeping, **0** stopped, **0** zombie

%Cpu(s): **0.1** us, **0.0** sy, **0.0** ni, **99.8** id, **0.0** wa, **0.0** hi, **0.0** si, **0.0** st

MiB Mem : **1705.8** total, **330.1** free, **918.0** used, **632.5** buff/cache

MiB Swap: **2048.0** total, **2048.0** free, **0.0** used. **787.8** avail Mem

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
2151	nivass	20	0	6308392	328484	131676	S	1.3	18.8	1:16.62	gnome-shell
1	root	20	0	175212	17636	11108	S	0.0	1.0	0:00.98	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_g
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-sync_
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slub_
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-netns
10	root	20	0	0	0	0	I	0.0	0.0	0:00.00	kworker/u32:0-events_unbound
11	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-mm_pe
12	root	20	0	0	0	0	I	0.0	0.0	0:00.01	kworker/u32:1-netns
13	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_kthre
14	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_rude_


```

top - 08:53:12 up 25 min,  1 user,  load average: 0.00, 0.03, 0.05
Tasks: 273 total,   1 running, 272 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.2 us,  0.0 sy,  0.0 ni, 99.7 id,  0.0 wa,  0.1 hi,  0.0 si,  0.0 st
MiB Mem :  1705.8 total,   330.1 free,   918.0 used,   632.5 buff/cache
MiB Swap:  2048.0 total,  2048.0 free,    0.0 used.   787.8 avail Mem

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
2151	nivass	20	0	6312552	328488	131676	S	2.0	18.8	1:17.30	gnome-shell
3284	nivass	20	0	765936	52196	40960	S	0.3	3.0	0:04.66	gnome-terminal-
3694	nivass	20	0	225928	4308	3380	R	0.3	0.2	0:00.53	top
1	root	20	0	175212	17636	11108	S	0.0	1.0	0:00.98	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_g
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-sync_
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slub_
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-netns
10	root	20	0	0	0	0	I	0.0	0.0	0:00.00	kworker/u32:0-events_unbound
11	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-mm_pe
12	root	20	0	0	0	0	I	0.0	0.0	0:00.01	kworker/u32:1-netns
13	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_kthre
14	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_rude_
15	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_trace
16	root	20	0	0	0	0	S	0.0	0.0	0:00.00	ksoftirqd/0
17	root	20	0	0	0	0	I	0.0	0.0	0:00.41	rcu_preempt
18	root	20	0	0	0	0	S	0.0	0.0	0:00.00	rcu_exp_par_gp_
19	root	20	0	0	0	0	S	0.0	0.0	0:00.01	rcu_exp_gp_kthr
20	root	rt	0	0	0	0	S	0.0	0.0	0:00.02	migration/0
21	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/0
23	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/0
24	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/1